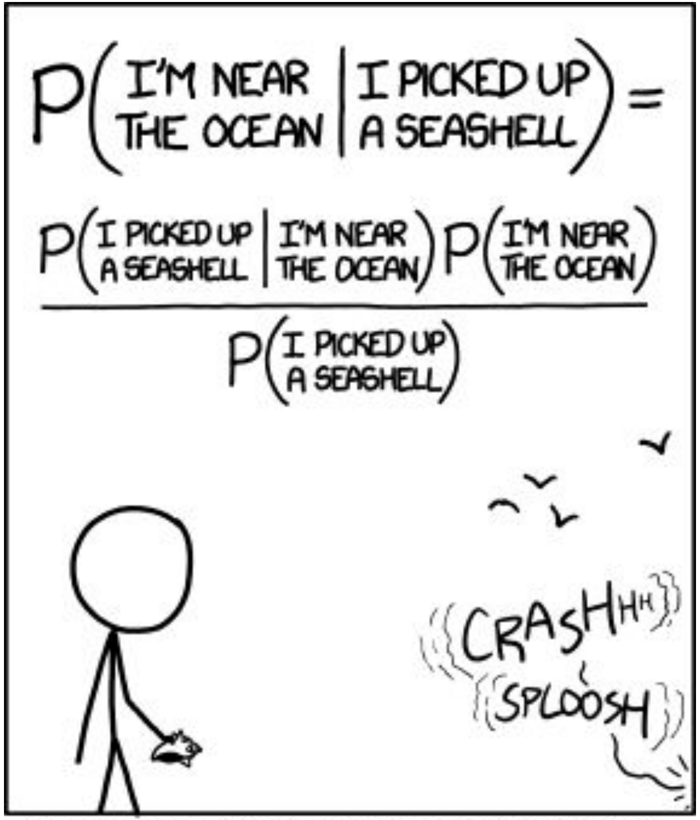
CS 159 Week 4:

Text Classification



STATISTICALLY SPEAKING, IF YOU PICK UP A SEASHELL AND *DON'T* HOLD IT TO YOUR EAR, YOU CAN PROBABLY HEAR THE OCEAN.

(https://xkcd.com/1236)

Last week:

A **language model** scores the probability of a given word sequence:

$$P(w_1, w_2, w_3, \ldots, w_k)$$

The simplest language model is the **bag of words model**, which assumes words are completely independent of each other:

$$P(w_1, w_2, w_3, \ldots, w_k) \approx P(w_1)P(w_2)P(w_3)\ldots P(w_k)$$

Each $P(w)$ can be estimated as the **frequency** of word type w

Last week:

- Word frequencies (and relative frequencies) aren’t only used to create bad language models…
- They are also extremely useful tools for studying language!
- Some examples:
  - Authorship attribution (each individual author has biases towards certain words)
  - Spam detection (some words are more common in spam emails)
  - Trend analysis (ever seen a word cloud?)
- Fun visualization of word frequencies: Google Books Ngram Viewer

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Doing useful things with word frequencies

What we can do with the tools we have so far:

Holmes, this text was written by Shakespeare.

Splendid! Then I can deduce that it has a high frequency of words like “thee” and “wherefore”.

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What we often want to do:

Holmes, this text contains a lot of words like “thee” and “wherefore”.

Splendid! Then I can deduce that it was likely written by Shakespeare.

A new formalism

Given a **document** $\mathrm{d} = \mathrm{w}_1, \mathrm{w}_2, \mathrm{w}_3, \ldots, \mathrm{w}_k;$

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And a set $C = \{c_1,\ldots,c_n\}$ of possible categories, or **classes**, that the document could belong to;

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$$P(c_i \mid d)$$

For each $c_i \in C$.

A new formalism: Classifiers

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We want to model the probability:

$$P(c_i \mid d)$$

For each $c_i \in C$. Such a model is called a **classifier**.


"QUICK! TO THE BAT- BOARD!
WATTERSON

Features can be anything!

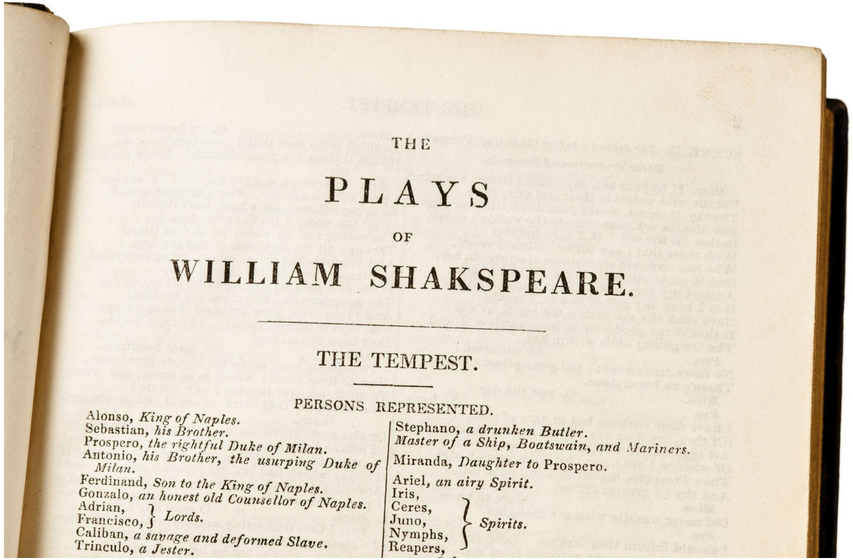
C = {cat, dog}

Features could include:

- Are there any yellow pixels?
- Is there anything that looks triangular? (This can be done using convolution)
- Are there any sharp edges? (This can be done using clever signal processing)

Features can be anything!

C = {Shakespeare, Chaucer, Keats}



Features could include:

- Is it in iambic pentameter?
- Is the relative frequency of "thou" greater than the relative frequency of "you"?
- Is the reading level of the text (computable with some simple heuristics) close to some expected value?

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LIES!

Text classification in practice

Most text classifiers don’t *only* use tokens as their features.

Tasks like spam detection and authorship attribution often *also* rely on hand-designed features (which may be derived from the text, or may come from metadata such as information about an email’s sender)

Let’s build some intuition for this!

Classification game

- Each of you has been handed a sheet of paper containing the subject line of an email, some metadata, and a label of whether it’s spam (look at your sheet but DON’T show it to anyone else!)
- Split into pairs. In each pair:
- Pick one partner to be the **questioner**, the other will be the **answerer.**
- The questioner’s task is to guess whether the answerer’s sheet contains a spam email, without seeing the actual text!
- To do this, they can ask any *specific* questions that *don’t* reveal the full text of the answerer’s email subject line (i.e., you can’t ask “what is the first word?”, etc.)
- Once the questioner feels confident, or have used up 20 questions, they will make a guess and the answerer will tell them if they were right.
- Then, switch roles!

Classification game: Debrief

Any strategies that worked particularly well? Any that didn't?

What was the smallest number of questions anyone was able to use to guess the correct answer?

The skills/intuitions you drew upon in this game may come in handy for Lab 3…